VNR Vignana Jyothi Institute of Engineering

Civil Engineering | Course Code: CE201

Surveying, Geomatics & Remote Sensing - Laboratory Manual & Practical Gu

Syllabus Focus: CE201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Fundamental Surveying Principles & Chain Survey

" Syllabus Scope: Principles of surveying, scales, error propagation, chain surveying, ranging, obstacles

Unit 2: Compass Surveying & Leveling Methods

" Syllabus Scope: Prismatic compass, magnetic declination, local attraction, differential leveling, rise

CE201: Surveying, Geomatics & Remote Sensing - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Theodolite Traversing & Curve Setting

" Vernier theodolite, horizontal and vertical angles, latitude and departure, Gales traverse table, simple cir

Unit 4: Tacheometry & Modern Total Station

" Stadia tacheometry principle, tangential tacheometry, electronic distance measurement EDM, Total Station com

Unit 5: Global Positioning Systems & GIS Analysis

" GPS segments space, control, user, satellite trilateration, pseudo-range errors, Geographic Information Syst

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Chain Traversing and Plotting of an Irregular Field Boundary with Obstacle Clearance

Lab 2: Differential and Profile Leveling with Longitudinal and Cross-Sectional Plotting

Lab 3: Measurement of Horizontal Angles by Repetition and Reiteration Methods using Vernier Theodolite

Lab 4: Topographical Feature Survey and Coordinate Mapping using Electronic Total Station

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* B.C. Punmia, Ashok K. Jain - Surveying Vol. I & II Laxmi Publications

* T.P. Kanetkar, S.V. Kulkarni - Surveying and Levelling Pune Vidyarthi Griha Prakashan

Authorized NoteVault Educational Document " Verified for Registered University Students